

RESEARCH ARTICLE

**Processing poultry manure with black soldier fly technology
lowers N₂O and CO₂ gas emissions from soil**

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Supplementary material

TABLE S1 Results of two-way ANOVA that included fertiliser treatments and time periods as categorical predictor variables of GHG flux in a full factorial design

Variable	Response	Sum Sq	Df	F value	P-value	
CO ₂ flux (kg CO ₂ -C ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	14.57	6	19.0	0.000	***
	Treatment*Time	18.01	18	7.8	0.000	***
	Residuals	14.05	110			
CH ₄ flux (kg CH ₄ -C ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	0.00	6	1.8	0.102	
	Treatment*Time	0.00	18	0.7	0.828	
	Residuals	0.00	112			
N ₂ O flux (kg N ₂ O-N ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	0.00	6	0.4	0.904	
	Treatment*Time	0.28	18	12.3	0.000	***
	Residuals	0.14	108			
Cumulative CO ₂ flux (kg CO ₂ -C ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	30.62	6	99.8	0.000	***
	Treatment*Time	12.61	18	13.7	0.000	***
	Residuals	5.73	112			
Cumulative CH ₄ flux (kg CH ₄ -C ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	0.00	6	1.2	0.311	
	Treatment*Time	0.00	18	1.0	0.443	
	Residuals	0.00	112			
Cumulative N ₂ O flux (kg N ₂ O-N ha ⁻¹ d ⁻¹)	(Intercept)	0.00	1	0.0	1.000	
	Treatment	0.00	3	0.0	1.000	
	Time	0.04	6	3.2	0.006	**
	Treatment*Time	0.79	18	20.2	0.000	***
	Residuals	0.23	108			

TABLE S2 Results of effects of treatments on N₂O flux through time using pairwise t-tests for each time point

	Time	Control				Frass				Manure				Urea			
N ₂ O flux	0	0.0000	±	0.0000		0.0000	±	0.0000		0.0000	±	0.0000		0.0000	±	0.0000	
(kg N ₂ O-N	6	0.0154	±	0.0036	a	0.0348	±	0.0068	b	0.1486	±	0.0238	c	0.0206	±	0.0079	ab
ha ⁻¹ d ⁻¹)	10	0.0158	±	0.0065	a	0.0490	±	0.0091	b	0.1304	±	0.0385	b	0.0172	±	0.0023	ac
	24	0.0288	±	0.0156		0.0520	±	0.0332		0.0368	±	0.0148		0.0226	±	0.0064	
	72	0.0150	±	0.0025	a	0.1646	±	0.0245	b	0.0448	±	0.0066	c	0.0308	±	0.0063	c
	168	0.0050	±	0.0035	a	0.0370	±	0.0067	b	0.1703	±	0.0303	c	0.0080	±	0.0023	a
	336	0.0044	±	0.0012	a	0.3633	±	0.0619	b	0.3743	±	0.0698	ab	0.0884	±	0.0337	a
Cumulative	0	0.0000	±	0.0000		0.0000	±	0.0000		0.0000	±	0.0000		0.0000	±	0.0000	
N ₂ O flux	6	0.0374	±	0.0080	a	0.0622	±	0.0117	a	0.3724	±	0.0334	b	0.0366	±	0.0079	a
(kg N ₂ O-N	10	0.0532	±	0.0087	a	0.1112	±	0.0165	b	0.5030	±	0.0470	c	0.0538	±	0.0081	a
ha ⁻¹ d ⁻¹)	24	0.0820	±	0.0215	ac	0.1634	±	0.0370	a	0.5402	±	0.0353	b	0.0762	±	0.0123	c
	72	0.0970	±	0.0235	a	0.3282	±	0.0501	b	0.5848	±	0.0334	c	0.1068	±	0.0118	a
	168	0.1020	±	0.0225	a	0.3652	±	0.0486	b	0.7507	±	0.0317	c	0.1150	±	0.0113	a
	336	0.1062	±	0.0216	a	0.7457	±	0.1476	b	1.0770	±	0.0352	c	0.2034	±	0.0332	a

TABLE S3 Results of effects of treatments on CO₂ flux through time using pairwise t-tests for each time point

	Time	Control			Frass			Manure			Urea		
CO ₂ flux	0	0.0000	±	0.0000	0.0000	±	0.0000	0.0000	±	0.0000	0.0000	±	0.0000
(kg CO ₂ -C	6	3.4100	±	0.4368	a	11.5276	±	0.3755	b	19.3358	±	3.6681	b
ha ⁻¹ d ⁻¹)	10	0.0068	±	0.0029	a	0.0210	±	0.0040	b	0.0556	±	0.0164	b
	24	2.9486	±	0.7443	a	22.8048	±	3.3554	b	20.7526	±	1.9150	b
	72	3.5876	±	0.8097	a	29.4908	±	6.7948	b	20.3146	±	2.5764	b
	168	3.2810	±	0.1303	a	9.4256	±	1.0954	b	29.4457	±	4.9320	ab
	336	4.1102	±	1.1973	ac	16.1163	±	6.4668	b	11.3950	±	3.9336	a
Cumulative	0	0.0000	±	0.0000	0.0000	±	0.0000	0.0000	±	0.0000	0.0000	±	0.0000
CO ₂ flux	6	6.2042	±	0.7678	a	23.3544	±	1.9865	b	76.1378	±	8.7631	c
(kg CO ₂ -C	10	6.2112	±	0.7672	a	23.3754	±	1.9871	b	76.1936	±	8.7686	c
ha ⁻¹ d ⁻¹)	24	9.1598	±	1.3287	a	46.1800	±	2.7638	b	96.9460	±	8.5494	c
	72	12.7470	±	1.7381	a	75.6712	±	8.9705	b	117.2604	±	9.4700	c
	168	16.0280	±	1.8559	a	85.0970	±	9.7610	b	141.5490	±	11.9463	c
	336	20.1384	±	1.0709	a	96.1177	±	23.1702	b	145.0470	±	8.4625	c

TABLE S4 Results of two-way ANOVA that included fertiliser treatments and time periods as categorical predictor variables of soil properties in a full factorial design

Variable	Response	Sum Sq	Df	F value	<i>P</i> -value	
DOC mg kg ⁻¹	(Intercept)	114.70	1	2111.9	0.000	***
	Treatment	5.46	3	33.5	0.000	***
	Time	1.46	6	4.5	0.000	***
	Treatment*Time	5.01	18	5.1	0.000	***
	Residuals	6.08	112			
NH ₄ -N mg kg ⁻¹	(Intercept)	56.76	1	143.4	0.000	***
	Treatment	21.77	3	18.3	0.000	***
	Time	41.79	6	17.6	0.000	***
	Treatment*Time	59.62	18	8.4	0.000	***
	Residuals	44.35	112			
NO ₃ -N mg kg ⁻¹	(Intercept)	92.31	1	475.5	0.000	***
	Treatment	0.17	3	0.3	0.829	
	Time	4.41	6	3.8	0.002	**
	Treatment*Time	14.27	18	4.1	0.000	***
	Residuals	21.75	112			
MC %	(Intercept)	46.31	1	12241.1	0.000	***
	Treatment	0.05	3	4.8	0.003	**
	Time	0.06	6	2.8	0.013	*
	Treatment*Time	0.10	18	1.4	0.129	
	Residuals	0.42	112			

TABLE S5 Results of effects of treatments on soil properties through time using pairwise t-tests for each time point

	Time	Control				Frass				Manure				Urea			
DOC mg kg ⁻¹	0	119.48	±	3.79	a	345.06	±	6.46	b	496.30	±	44.36	c	324.60	±	13.14	d
	6	113.92	±	4.06	a	340.98	±	37.64	bc	437.48	±	58.82	c	274.12	±	10.17	b
	10	189.08	±	23.53	a	469.04	±	27.92	b	493.92	±	59.42	b	235.82	±	9.94	ac
	24	132.46	±	23.71	a	308.02	±	21.17	b	411.38	±	111.78	b	166.28	±	8.58	ac
	72	95.22	±	3.95	a	196.60	±	15.71	b	332.86	±	54.28	c	130.88	±	13.78	d
	168	153.72	±	16.80	a	200.18	±	9.45	b	457.63	±	92.55	c	140.48	±	9.96	a
	336	155.48	±	6.88	ab	191.50	±	1.99	c	196.17	±	26.62	bc	147.56	±	5.02	a
NH ₄ -N mg kg ⁻¹	0	30.80	±	5.72	a	126.28	±	6.31	b	324.90	±	35.84	ac	33.26	±	8.66	d
	6	33.02	±	2.66	a	145.08	±	16.33	b	330.90	±	45.88	c	123.18	±	4.38	b
	10	30.98	±	5.09	a	186.88	±	7.73	b	355.16	±	39.17	c	201.84	±	7.79	b
	24	23.04	±	9.20	a	124.28	±	25.10	b	377.78	±	90.09	c	547.60	±	27.51	c
	72	10.84	±	2.33	a	140.32	±	23.23	b	516.46	±	168.76	c	510.72	±	48.78	c
	168	0.36	±	0.09	a	37.70	±	7.33	b	682.93	±	111.52	c	145.66	±	5.25	d
	336	16.54	±	14.39	a	13.47	±	6.00	a	63.87	±	9.70	b	64.34	±	16.65	ab
NO ₃ -N mg kg ⁻¹	0	74.66	±	8.83	ab	62.04	±	8.09	ab	75.10	±	3.53	a	73.98	±	2.11	b
	6	61.90	±	2.79		62.62	±	4.81		104.88	±	23.93		64.00	±	5.80	
	10	66.88	±	9.52		62.98	±	4.80		104.12	±	23.42		59.72	±	6.36	
	24	59.00	±	8.73	ab	62.96	±	9.66	ab	52.06	±	4.94	a	74.36	±	6.54	b
	72	95.42	±	4.85	a	28.46	±	14.57	b	39.56	±	11.59	b	103.62	±	6.94	a
	168	125.86	±	3.68	a	87.86	±	17.25	b	90.47	±	17.82	ab	259.32	±	2.83	c
	336	168.58	±	42.53	a	95.60	±	43.88	a	118.87	±	36.82	a	303.08	±	42.01	b
MC %	0	20.26	±	1.88	a	17.58	±	0.18	b	17.64	±	0.29	b	17.58	±	0.29	b
	6	17.18	±	0.30		17.14	±	0.17		17.08	±	1.65		17.52	±	0.23	
	10	17.60	±	0.23	a	18.68	±	0.28	bc	19.16	±	0.12	b	18.28	±	0.10	c
	24	18.26	±	0.25	ab	18.36	±	0.26	ab	19.08	±	0.34	a	17.88	±	0.30	b
	72	18.56	±	0.18		18.36	±	0.15		19.06	±	0.39		18.24	±	0.26	
	168	18.18	±	0.11	a	18.70	±	0.23	a	18.63	±	0.54	ab	17.62	±	0.18	b
	336	17.74	±	0.22	a	18.43	±	0.41	b	18.63	±	0.35	b	18.30	±	0.13	ab

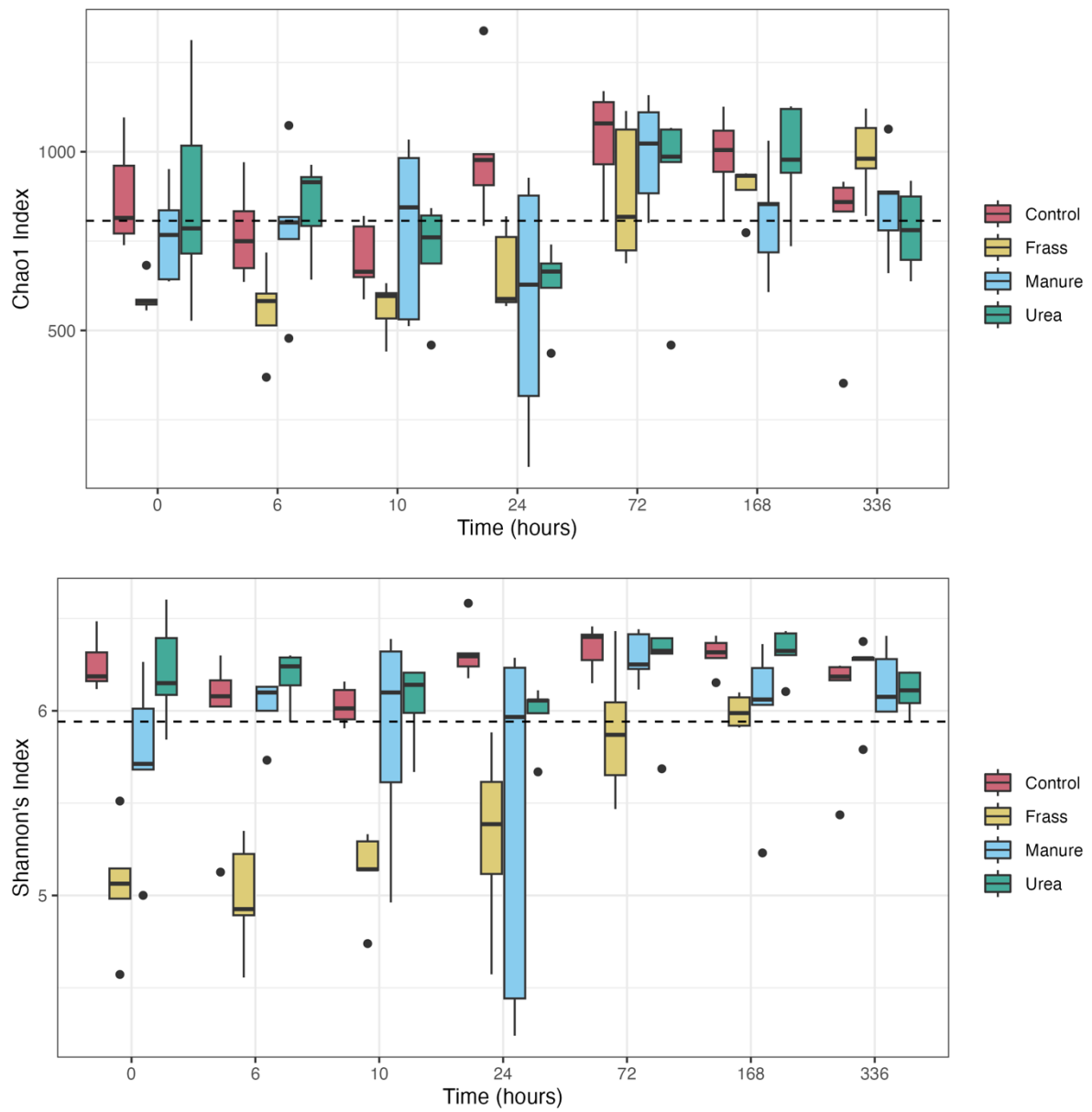


FIGURE S1 Microbial alpha diversity among treatments for each time point.